MILITARY SPECIFICATION

TRANSMISSION, POWER SHIFT, COMMERCIAL MODIFIED

This specification is for use by US Army Tank-Automotive Materiel Readiness Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Power Shift Transmission, as modified for use in M746 military vehicle.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

P-C-437	- Cleaning Compound, High Pressure (Steam) Cleaner.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-L-2104	- Lubricating Oil, Internal-combustion Engine, Tactical Service.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Materiel Readiness Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2520

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:21Z
Check the source to verify that this is the current version before use.

MIL-E-22118	- Enamel, Electrical-insulating.
MIL-L-46167	- Lubricating Oil, Internal-combustion Engine, Arctic.
MIL-E-62146	- Engine, Diesel: 12 Cylinder, V-type, 600 HP, 2 Cycle Liquid Cooled, Turbo-supercharged.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Finishing Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1184	- Electrical Components for Automotive Vehicles, Waterproofness Tests.

DRAWINGS

ARMY

CPR 102106	- Engine Assembly.
CPR 102393	- Hydraulic Torque Converter with Transmission.
CPR 109517	- Transmission Range Selector Assembly.

QUALITY ASSURANCE REQUIREMENTS (QAR's)

MASTER LIST

CPR 102393	- Master List of Quality Assurance Requirements for Main Assembly - Hydraulic Torque Converter with Transmission.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article (initial production) transmission. The transmission selected as first production shall be fully representative of transmissions proposed to be furnished under the contract, with all current modifications included in and on the selected transmission. If submitted transmission meets all requirements of this specification, no modifications shall be applied to subsequent transmissions produced under contract without prior Government approval (see 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards and drawings. Material shall be free of defects which adversely affect performance or serviceability of the finished product (see 6.5).

3.2.1 Ozone resistance. When rubber components are required to be ozone resistant, the degree of resistance shall be as specified in applicable specifications or drawings.

3.2.2 Fungus resistance. The transmission assembly, including accessories, shall meet the fungus requirements of method 508, procedure 1, of MIL-STD-810.

3.2.3 Oil. The transmission shall be filled to operating level with oil conforming to grade 10 of MIL-L-2104 when operating in ambient air temperature above 0°F. Oil conforming to MIL-L-46167 shall be used when operating in ambient air temperature below 0°F. Oil level shall be maintained within the operating range on the oil level indicator.

3.3 Design and construction.

3.3.1 Transmission. The transmission assembly shall be in accordance with CPR 102393, and interface and be compatible with engine CPR 102106. The transmission shall have a torque converter, with automatic lock-up, retarder, and transfer case integrally assembled. The transmission shall interface and transmit the engine power and torque as specified in MIL-E-62146.

3.3.1.1 Interchangeability of parts. Component assemblies and parts of the transmission shall be so constructed that they may be installed, replaced and adjusted without requiring modification when used with other transmissions provided under the same part number. If some parts must be furnished in matched sets, the contractor shall provide a list giving part number, part nomenclature and tolerance ranges of sets provided. This list must be approved by the Government prior to shipment of any transmission.

3.3.1.2 Accessories and equipment. Unless otherwise specified, all accessories and equipment shall be installed and properly adjusted. All electrical accessories and equipment including wiring and electrical connections shall conform to the applicable requirements of MIL-STD-1184.

3.3.1.3 Shifting controls. Solenoid selector valves shall be furnished integral with electric selector controls as specified in CPR 109517. All solenoid ground connections shall be coated in accordance with MIL-E-22118.

3.3.1.4 Seals. With the transmission operating under all environmental conditions specified herein, seals shall restrict entrance of all foreign material into the bearings, which are exposed to contamination during these operations. Seals shall prevent water contamination of lubricants and lubricant leakage.

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:21Z
Check the source to verify that this is the current version before use.

3.4 Performance.

3.4.1 Oil.

3.4.1.1 Temperature range. The converter oil temperature shall not exceed 275°F during each transmission evaluation sequence and production run.

3.4.1.2 Oil level. Transmission oil level shall be maintained within the operating range on the oil level indicator. The oil level shall be between "full" and "add oil" on the transmission, while operating in neutral range at 700 to 900 rpm input speed. The oil level shall be maintained by adding or removing oil.

3.4.1.3 Leakage. During any evaluation sequence or at completion of all evaluations, the transmission shall not evidence leakage in excess of a seep as defined in 6.4.

3.4.1.4 Oil pressure. The transmission oil pressure, with the "out" oil temperature of the converter stabilized at 200°F to 230°F, shall conform to table I. All pressures measured in psi.

TABLE I. Hydraulic pressures.

Range	Input speed	Mode	Main	Lube	Converter	Pitot	Lock-up clutch	Oil cooler flow
R 1	1100 ± 20	Converter	160 min.	6-20	45-60			22.5 gpm min.
N 2,3,4,5	1100 ± 20	Converter	160 min.	6-20	45-50			22.5 gpm min.
R 1	2190 ± 20	Converter	205-230	10-20	50-60			52.5 gpm min.
N 2,3,4,5	2190 ± 20	Converter	170-195	10-20	50-60			52.5 gpm min.
N 2,3,4,5	2190 ± 20	Lockup	170-195	10-20	50-60	40-50	165-195	52.5 gpm min.

3.4.2 Lock-up. The automatic clutch shall engage automatically at 2035 ± 50 rpm and disengage at 1735 ± 50 rpm turbine speed in N 2,3,4 and 5 ranges.

3.4.3 Retarder. The retarder shall provide horsepower absorption as follows: With the transmission in neutral, and the converter manually engaged in lock-up, the retarder shall absorb a minimum of 30 horsepower at a minimum input speed of 725 ± 10 rpm. Minimum oil to cooler flow rate shall be 11.5 gallons per minute (gpm) at 690 ± 20 rpm.

3.4.3.1 Dual range. With the transmission engaged in two ranges and the output shaft stalled, the transmission shall absorb a minimum of 14 horsepower at an input speed of 725 ± 10 rpm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

3.4.3.2 Biasing differential. With transmission in first range at 2190 ± 20 rpm input, and 40 psi air pressure applied to differential clutch control valve, clutch pressure at the valve body shall be 130 ± 10 psi. With driving motor off, but with pumps on, differential clutch engagement shall be accomplished by holding one input shaft and turning the other. Clutch shall release at zero air pressure and engage at 40 pounds air pressure.

3.4.4 Transmission friction. With the "out" oil temperature of the converter stabilized at 200°F to 230°F, at 2190 ± 20 input rpm and no load on the output shaft, the transmission frictional losses shall not exceed that specified in table II. Frictional torque losses shall be measured in lb. ft.

TABLE II. Transmission friction losses.

Reverse	Neutral	1st	2nd	3rd	4th	5th
79	74	77	77	83.5	100	130

3.4.5 Transmission break-in run. Each transmission shall be subjected to a break-in run by the contractor (see 4.3.4).

3.4.6 Slopes. The transmission shall demonstrate the performance specified herein when operated at 1750 ± 50 rpm for not less than 30 minutes in each direction with full, and one gallon low, oil level conditions in the sump on slopes up to 30 percent (16.7 degrees) in all directions. As a result of said operation, no evidence of faulty lubrication, cooling, oil supply, leakage, excessive noise or other malfunction shall occur.

3.4.7 Environmental operation.

3.4.7.1 High temperature operation. The transmission, without external aids and with provisions for oil cooling, shall operate through all gear ranges at an ambient temperature of 120°F.

3.4.7.2 Low temperature operation. The transmission, without external load, shall operate through all gear ranges without benefit of solar radiation after being cold soaked to an ambient temperature of minus 65°F.

3.4.7.3 Submersion. All electrical components, connections, gaskets, seals, and joints shall prevent water seepage into the interior of the converter, transmission and transfer case assembly when submerged in water to centerline of input shaft for a period of 30 minutes. After removal from water, mechanical and hydraulic systems shall perform as specified herein.

3.4.7.4 Steam and water jet cleaning. The transmission as installed in vehicle, or a simulated mounting with protection to the input and output shafts, shall withstand cleaning with high pressure steam and cleaning compound conforming to P-C-437 and water jet, without deterioration of seals,

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

water leakage past seals or gaskets, or other defects. Paint removal shall not be considered a defect.

3.5 Exterior surface treatment. All exposed surfaces of the transmission and its components shall be cleaned, painted, or treated for corrosion resistance as specified on the applicable drawings, or if not so specified, in accordance with applicable provisions of MIL-STD-193.

3.6 Identification markings. The complete transmission and each major component shall be marked as specified on applicable drawings or, if not so specified, in accordance with MIL-STD-130.

3.6.1 Name plate. Unless otherwise specified on the applicable drawing, the name plate, data and instruction plates shall conform to the applicable provisions of MIL-P-514.

3.7 Log sheet. The contractor shall maintain a log sheet of each transmission tested as specified herein. The form of the log sheet and number of legible copies to be furnished to the Government shall be as approved by the procuring activity (see 6.2).

3.8 Workmanship. The workmanship shall be of such quality as to assure that all transmissions procured under contract shall meet performance requirements specified herein. Electric, and lubrication lines shall be located in a manner so as to prevent possible damage by rubbing on adjacent lines or appendages. All parts and assemblages shall be free of burrs, sharp edges or any condition which may present a safety hazard to operating and maintenance personnel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system. The contractor shall provide and maintain an effective inspection and quality assurance system, acceptable to the Government, covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed, provided that an equivalent control is included in the approved quality assurance procedure. The contractor shall notify the Government of and obtain approval for any change to the submitted procedure that might affect the degree of control required by this specification, or other applicable documents referenced herein.

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:21Z
Check the source to verify that this is the current version before use.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and Government product inspection to measure quality of product offered for acceptance. Deviation from prescribed or agreed-upon procedures, or instances of poor practices which might have an adverse effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until corrective action has been made, or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Materials and qualified products. The contractor's records shall be reviewed to determine conformance to 3.2 and 3.2.1.

4.1.4 Parts and components. Parts, components and assemblies shall be inspected in accordance with Master List of Quality Assurance Requirements CPR 102393 (see 2.1).

4.2 Classification of inspections. Inspections specified herein are classified as follows:

a. First production transmission inspection (see 4.4).
b. Quality conformance inspection (see 4.7).

4.3 Inspection conditions.

4.3.1 Input power source. The input power supply shall conform to MIL-E-62146, or a suitable equivalent power source.

4.3.2 Linkage controls. Separate linkage controls shall be provided for each of the following:

a. Engine throttle.
b. Range selector shift lever.

4.3.3 Instrumentation. Instrumentation shall be furnished to measure the following:

a. Input speed - rpm.
b. Output speed - rpm.
c. Main oil pressure - psi.
d. Transmission oil pressure - psi.
e. Converter out oil temperature.
f. Oil flow - gpm.
g. Input torque - pound feet (lb ft).
h. Output torque - pound feet (lb ft).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

4.3.4 Break-in run. To determine conformance to 3.4.5, each transmission shall be operated for not less than 15 minutes prior to production tests.

4.3.5 Log sheet. To determine conformance to 3.7, a reproducible test log sheet shall be maintained incorporating data taken during evaluation of each transmission (see 6.2).

4.4 First production transmission inspection.

4.4.1 In-process inspection. During fabrication of the first production transmission, an in-process inspection will be conducted by the Government to evaluate conformance of materials and workmanship to specified requirements. Inspection shall be made at contractor's facility prior to application of primer and paint. Processing, welding procedures, quality system and inspection records will be reviewed and evaluated during this inspection.

4.4.2 Repair of defects. Defects found as a result of the foregoing inspection shall be corrected by the contractor at no cost to the Government. Failure of the contractor to promptly correct defects shall be cause for suspension of acceptance of transmissions, until corrective action has been accomplished.

4.4.3 Transmission disposition. After completion of inspection, the first production transmission shall remain at the manufacturing facility, as a production sample, and be the last transmission shipped on the contract. Transmission may be released at the discretion of the Government. The contractor shall service and maintain the transmission during this period.

4.4.4 Final approval and acceptance. Final approval and acceptance by the Government of the first production transmission shall be withheld until acceptance testing of the second transmission produced has been completed and a final determination has been made regarding conformity of the transmission to requirements. This shall include, but not be limited to, workmanship and materials.

4.5 First article (initial production) test. To determine conformance to Section 3 (inclusive), after completion of the first production inspection, another transmission shall be selected and subjected to examination specified in 4.7.1. Subsequent to examination, transmission shall be subjected to all testing specified in table IV. The contractor shall furnish repair parts, as required, to support the tests at no cost to the Government.

4.5.1 Failure. Failure of the transmission to pass any specified test, or any defects of a workmanship or materials nature found during, or as a result of, the tests shall be cause for rejection of the transmission. Further, the Government may refuse to continue acceptance of production transmissions until evidence has been provided by the contractor that corrective action has been taken to eliminate such defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

4.6 Conformance verification.

4.6.1 Temperature range test. To determine conformance to 3.4.1.1, the transmission shall be operated as specified, and observed for meeting requirements.

4.6.2 Oil level test. To determine conformance to 3.4.1.2, the transmission shall be operated as specified, and observed for meeting requirements.

4.6.3 Leakage test. To determine conformance to 3.4.1.3, the transmission shall be visually inspected at various points during test operation, and thoroughly inspected at the end of the test operation for indications of oil leaks.

4.6.4 Oil pressure test. To determine conformance to 3.4.1.4, the "out" oil temperature of the converter shall be stabilized at 200°F to 230°F, and all oil pressures shall fall within the limits shown in table I.

4.6.5 Lockup test. To determine conformance to 3.4.2, the transmission input speed shall be gradually increased to 2190 ± 20 rpm and engagement point observed. Subsequently motor shall be shut off, and lockup disengagement point observed as motor coasts to stop.

4.6.6 Retarder test. To determine conformance to 3.4.3, with the transmission operating as specified, the retarder shall be observed for meeting the absorption requirements.

4.6.7 Dual range test. To determine conformance to 3.4.3.1, with the transmission operating as specified, the transmission shall meet the absorption requirements specified.

4.6.8 Bias differential test. To determine conformance to 3.4.3.2, with pressure applied to transfer case as specified, the bias differential shall meet the requirements specified.

4.6.9 Transmission friction test. To determine conformance to 3.4.4, the transmission, operating as specified, shall be observed for maximum friction losses.

4.6.10 Slope test. To determine conformance to 3.4.6, the transmission shall be mounted in a fixture and operated as specified to simulate slope operation of 30 percent in all directions.

4.6.11 High and low temperature operation and storage test. To determine conformance to 3.4.7.1 and 3.4.7.2, the transmission shall be operated and stored as specified and observed for meeting requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

4.6.12 Submersion test. To determine conformance to 3.4.7.3, after submersion operation is completed, the specified items shall be examined for water content in accordance with 3.3.1.4.

4.6.13 Steam and water jet cleaning test. To determine conformance to 3.4.7.4, the initial production transmission shall be subjected to cleaning as specified. Steam and water jet cleaning as it pertains to this transmission is defined as follows: The jet is applied perpendicular to the surface to be cleaned at a minimum distance of one foot from the surface, and cleaning is accomplished at a minimum rate of one square foot per minute. The jet pressure shall be not less than 100 psi nor more than 110 psi. The seals and gaskets shall be examined for damage and leaks (see 3.3.1.4).

4.6.14 Log sheet. To determine conformance to 3.7, a reproducible test log sheet shall be maintained incorporating data taken during evaluation and testing of each transmission.

4.5.15 Preservation, packaging and transmission processing inspection. The transmission and equipment shall be inspected, prior to shipment, to determine conformance to Section 5 of this specification.

4.7 Quality conformance inspection.

4.7.1 Inspection provisions. Inspection shall consist of examination to determine conformance to applicable drawings (see 3.3) and this specification. Examination shall be performed against the classification of defects and with the acceptable quality levels (AQL's) specified in 4.7.2 and table III.

4.7.2 Acceptable quality level. Each transmission shall be examined for conformance to the following acceptable quality levels (AQL's) on the basis of percent defective:

Classification	AQL
Major	2.5
Minor	4.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

TABLE III. Classification of defects.

Categories	Defect	Method of inspection
Major		
101	Dimensions affecting interchangeability not within tolerance (see 3.3.1 and 3.3.1.1)	Gage or scale
102	Malfunction of control levers or valves (see 3.3.1.3)	Functional/ visual
103	Oil leakage (see 3.4.1.3)	Visual
104	Seal leakage (see 3.3.1.4)	Visual
105	Faulty workmanship (see 3.8)	Visual
Minor		
201	Dimensions not affecting interchange-ability, not within tolerance (see 3.3.1.1)	Gage or scale
202	Improper application or color of paint (see 3.5)	Visual
203	Improper or illegible marking (see 3.6)	Visual
204	Appearance - scratched, chipped, pitted or burred (see 3.8)	Visual
205	Dirt or foreign matter plugging lines, screens, or oil filters (see 3.8)	Visual

4.8 Classification of tests. Classification of tests shall be as follows:

a. Acceptance tests (see 4.8.1).
b. Control tests (see 4.8.2).

4.8.1 Acceptance tests (100 percent). To determine conformance to Section 3 (inclusive), each transmission shall be examined as specified in 4.7.1 and subjected to all acceptance tests specified in table IV. Transmissions shall be completely assembled and serviced with specified lubricants.

NOTE: The sequence of tests may be altered to be compatible with contractor's test facilities and practices.

4.8.1.1 Test failure. If a transmission fails to pass any acceptance test specified herein, the Government will withhold acceptance of subsequent transmissions until evidence has been provided by the contractor that corrective action has been taken.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

4.8.1.2 Re-submission. Rejected transmissions may be re-submitted for acceptance after defects have been repaired and evidence specified in 4.8.1.1 has been received.

TABLE IV. Test requirements.

	Requirement	Test	IPT	Accept-ance	Con-trol
Temperature range test	3.4.1.1	4.6.1	X	X	X
Oil level test	3.4.1.2	4.6.2	X	X	X
Leakage test	3.4.1.3	4.6.3	X	X	X
Oil pressure test	3.4.1.4	4.6.4	X	X	X
Lock-up test	3.4.2	4.6.5	X	X	X
Retarder test	3.4.3	4.6.6	X	X	X
Dual range test	3.4.3.1	4.6.7	X	X	X
Bias differential test	3.4.3.2	4.6.8	X	X	X
Transmission friction test	3.4.4	4.6.9	X	X	X
Slopes test	3.4.6	4.6.11	X		
High and low temperature	3.4.7.1	4.6.12	X		
operation and storage test	3.4.7.2	4.6.12	X		
Submersion test	3.4.7.3	4.6.13	X		
Steam and water jet cleaning test	3.4.7.4	4.6.14	X		
Production break-in run	3.4.5	4.3.4	X	X	X

4.8.2 Control tests. Transmission shall be selected at the rate of one of the first 35 units produced, and one out of each of the next 100 units thereafter. Transmissions shall be identified as to production period. The transmission, filled with oil specified herein, shall be mounted on a test stand (see 4.3), coupled to a motor unit at the input end, and a power absorption unit at the output end. Each transmission selected shall be examined for the defects specified in table III, and subjected to the 48 hour run specified in table V. After completion of the 48 hour run, transmission shall meet the control test requirements specified in table IV.

TABLE V. 48-Hour run.

Test cycle 1/	Shift range	Hours 1/	Output rpm	Output min.	Torque max.
1	1	.25	367	5800	6210
2	1	.5	514	4000	4310
3	2	.5	918	2200	2460
4	2	1.5	435	4350	4600
5	3	.5	1265	1550	1760
6	3	1.5	600	3100	3330
7	4	.5	1780	1200	1350
8	4	2.0	850	2200	2345
9	5	1.5	2100	950	1080
10	5	2.5	1780	850	973

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:21Z
Check the source to verify that this is the current version before use.

TABLE V. 48-Hour run. - Continued

Test cycle 1/	Shift range	Hours 1/	Output rpm	Output min.	Torque max.
11	R	.25	367	5220	5589
12	R	.5	514	3600	3951

1/ Repeat sequence 4 times.

4.8.2.1 Tear-down inspection. Subsequent to foregoing tests, the transmission shall be disassembled and the following examinations performed:

a. Gear wear
b. Galling
c. Bearing wear
d. Scoring
e. Chips and dirt

4.8.2.2 Control test failure. Failure of the transmission to conform to the requirements specified, or any major defect or workmanship or material found as a result of the test, shall be cause for rejection, and for refusal to accept subsequent transmissions. Defects found during, or as a result of, the test shall be prima facie evidence that transmissions previously accepted are similarly defective, unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly defective. Defects on all transmissions shall be corrected by the contractor at no cost to the Government. In the event of a failure, another transmission from the same production period shall be subjected to control testing.

4.8.2.3 Corrections and reassembly. Should evidence be found of a condition requiring replacement of components, such replacement shall be made by the contractor prior to reassembly. The transmission, with replaced components, shall then be subjected to acceptance tests (see 4.8.1).

4.8.2.4 Inspection and test results. The contractor shall provide inspection test results for all control tests. All data shall be obtained under stabilized operating conditions.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordacne with the applicable packaging data sheet as furnished at the time of invitation for bid for the level of packaging and level of packing specified by the procuring activity (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The transmission covered by this specification is intended for production use, as spares, or as replacement in military transport vehicles.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Number of log sheets to be furnished (see 3.7).
c. Selection of the applicable level of packaging, and level of packing (see 5.1).

6.3 First article. First article samples shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, tests and approval of the first article (see 3.1).

6.4 Definitions. The following definitions shall be used for the classification of defects for leaks:

a. Weep - Any evidence of fluid beyond the seal.
b. Seep - Any evidence of fluid beyond the seal that does not result in formation of a droplet.
c. Droplet - Any evidence of fluid beyond the seal that results in formation of a non-falling droplet.
d. Drip - Any evidence of fluid beyond the seal where droplets form and fall.

6.5 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:21Z
Check the source to verify that this is the current version before use.

6.6 Changes from previous issue. Asterisks are not used in this revision, to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No. 2520-A197

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:21Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-62148A(AT)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:21Z
Check the source to verify that this is the current version before use.